

```

clc; clear; close all;
% Task 1a: 4 point moving filter

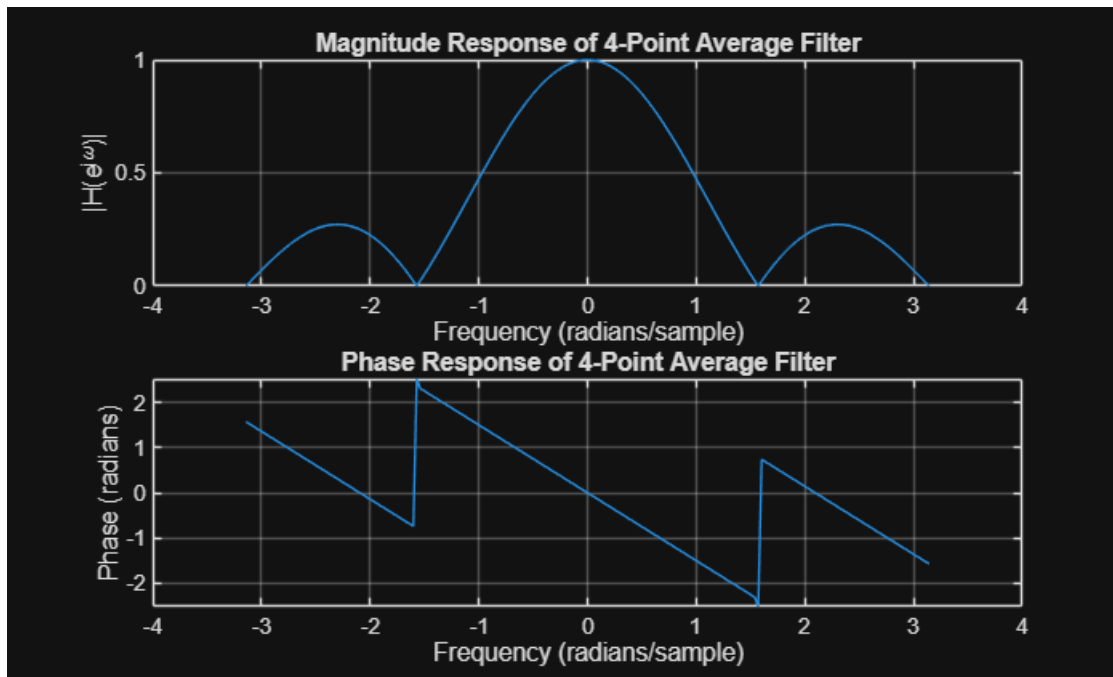
K = 4;
bb = ones(1, K) / K;
ww = -pi : (pi/100) : pi; % Frequency range

H = freqz(bb, 1, ww); % Frequency response

figure;
subplot(2,1,1);
plot(ww, abs(H));
grid on;
title('Magnitude Response of 4-Point Average Filter');
xlabel('Frequency (radians/sample)');
ylabel('|H(e^{j\omega})|');

subplot(2,1,2);
plot(ww, angle(H));
grid on;
title('Phase Response of 4-Point Average Filter');
xlabel('Frequency (radians/sample)');
ylabel('Phase (radians)');

```



Task 1b: Smoothing Effect

The magnitude response of the 4-point moving average filter shows a low-pass characteristic. Low-frequency components pass through the filter with little attenuation, while high frequency components are significantly

reduced. Since rapid variations or noise in signals correspond to high frequencies, their attenuation results in a smoothing effect on the signal.

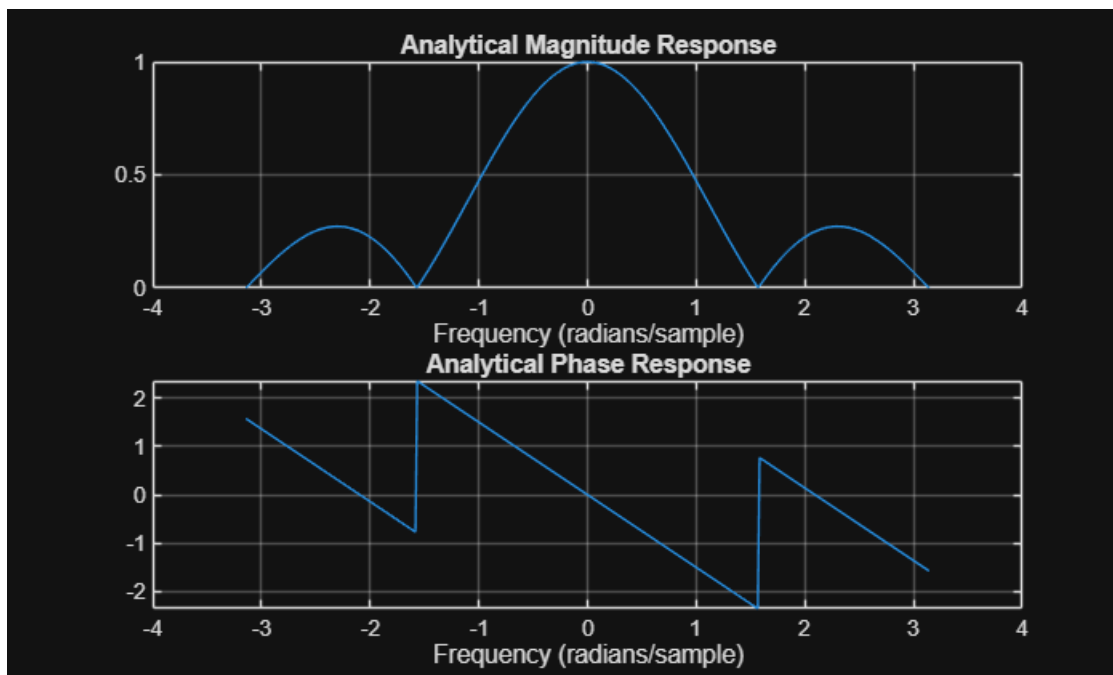
```
% Task 1c: Analytical implementation
w_analytical = linspace(-pi, pi, 400);

H_analytical = ((2*cos(0.5*w_analytical) + 2*cos(1.5*w_analytical)) / 4) ...
    .* exp(-1j * 1.5 * w_analytical);

mag_H = abs(H_analytical);
phase_H = angle(H_analytical);

figure;
subplot(2,1,1);
plot(w_analytical, mag_H);
title('Analytical Magnitude Response');
grid on;
xlabel('Frequency (radians/sample)');

subplot(2,1,2);
plot(w_analytical, phase_H);
title('Analytical Phase Response');
grid on;
xlabel('Frequency (radians/sample)');
```



```
% Task 1d: Effect of different K

ww = linspace(-pi, pi, 1000); % Frequency axis
K_values = [2, 4, 8, 16]; %filter lengths
```

```

figure;
hold on;

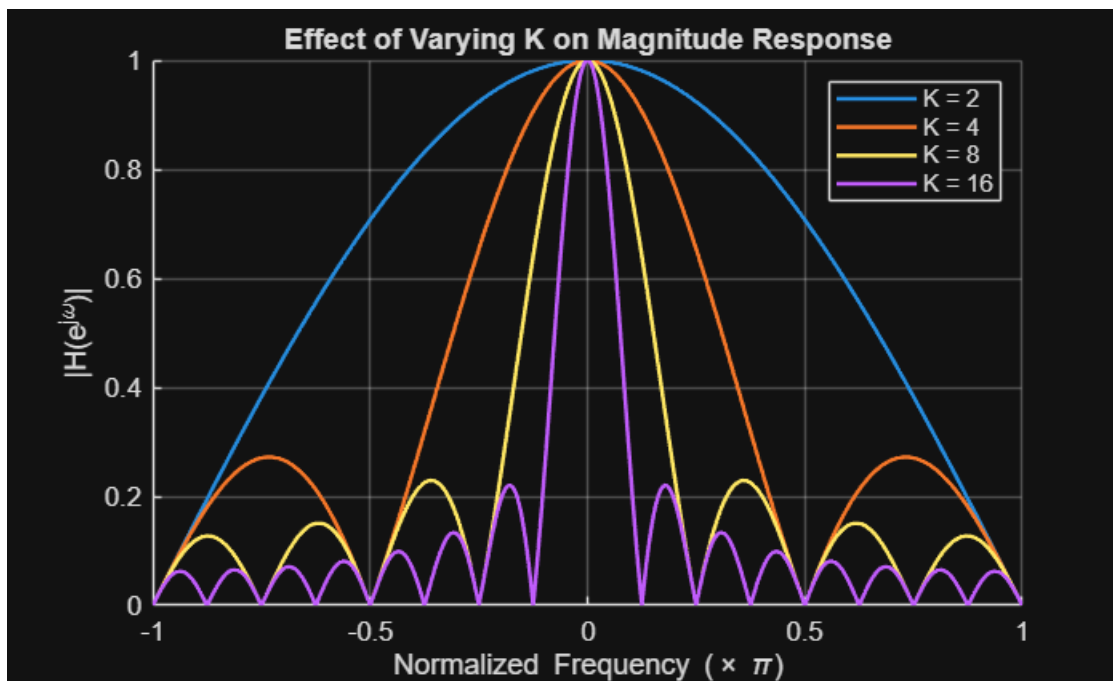
for i = 1:length(K_values)
    K = K_values(i);
    bb = ones(1, K) / K; % Moving filter

    H = freqz(bb, 1, ww); %frequency response

    plot(ww/pi, abs(H), 'LineWidth', 1.5, ...
        'DisplayName', ['K = ', num2str(K)]);
end

grid on; ghd
title('Effect of Varying K on Magnitude Response');
xlabel('Normalized Frequency (\times \pi)');
ylabel('|H(e^{j\omega})|');
legend;
hold off;

```



```

% Task 2 (a & b): Band-pass filter

```

```

wc = 0.44 * pi;
L_values = [10, 20, 40];
ww = linspace(-pi, pi, 1000);

figure;
for i = 1:length(L_values)
    L = L_values(i);

```

```

n = 0:L-1;

hh = (2/L) * cos(wc * n); % Impulse response
H = freqz(hh, 1, ww);

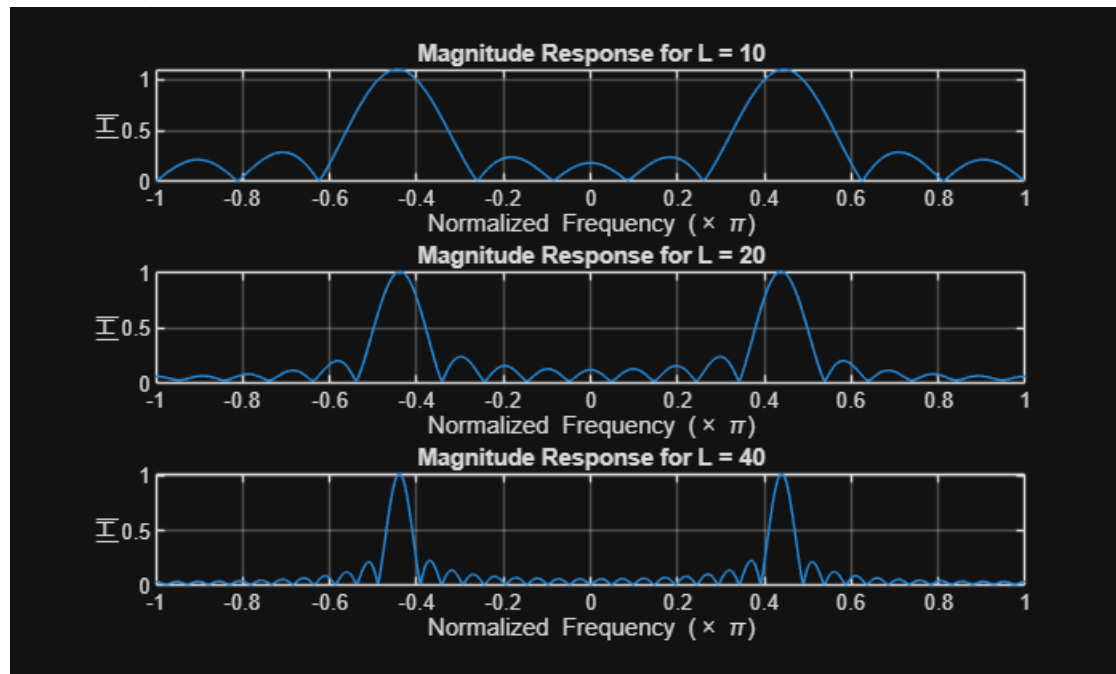
subplot(3,1,i);
plot(ww/pi, abs(H));
grid on;
title(['Magnitude Response for L = ', num2str(L)]);
xlabel('Normalized Frequency (\times \pi)');
ylabel('|H|');

% Passband width calculation
H_max = max(abs(H));
indices = find(abs(H) >= (H_max / sqrt(2)));
width = ww(indices(end)) - ww(indices(1));

fprintf('For L = %d, Passband Width = %.4f radians\n', L, width);
end

```

For L = 10, Passband Width = 3.3020 radians
 For L = 20, Passband Width = 3.0252 radians
 For L = 40, Passband Width = 2.8869 radians



```

% Task 2(c): Filter design
wc = 0.44 * pi;
L = 66; % chosen after testing
n_idx = 0:L-1;
h_bp = (2/L) * cos(wc * n_idx);

ww = linspace(-pi, pi, 1000);
H = freqz(h_bp, 1, ww);

```

```
H_max = max(abs(H));
```

```
% Check attenuation at unwanted frequencies
```

```
[~, idx1] = min(abs(ww - 0.3*pi));
```

```
[~, idx2] = min(abs(ww - 0.7*pi));
```

```
fprintf('Attenuation at 0.3pi = %.4f\n', abs(H(idx1))/H_max);
```

```
Attenuation at 0.3pi = 0.0593
```

```
fprintf('Attenuation at 0.7pi = %.4f\n', abs(H(idx2))/H_max);
```

```
Attenuation at 0.7pi = 0.0440
```

```
% Task 2d: Testing with input signal
```

```
n = 0:200;
```

```
x = 10*cos(0.3*pi*n) + 40*cos(0.44*pi*n - pi/3) + 20*cos(0.7*pi*n - pi/4);
```

```
y = filter(h_bp, 1, x);
```

```
% FFT Analysis
```

```
N_fft = length(y);
```

```
Y_fft = fftshift(fft(y));
```

```
% Frequency axis
```

```
f_axis = (-N_fft/2 : N_fft/2-1) * (2/N_fft);
```

```
figure;
```

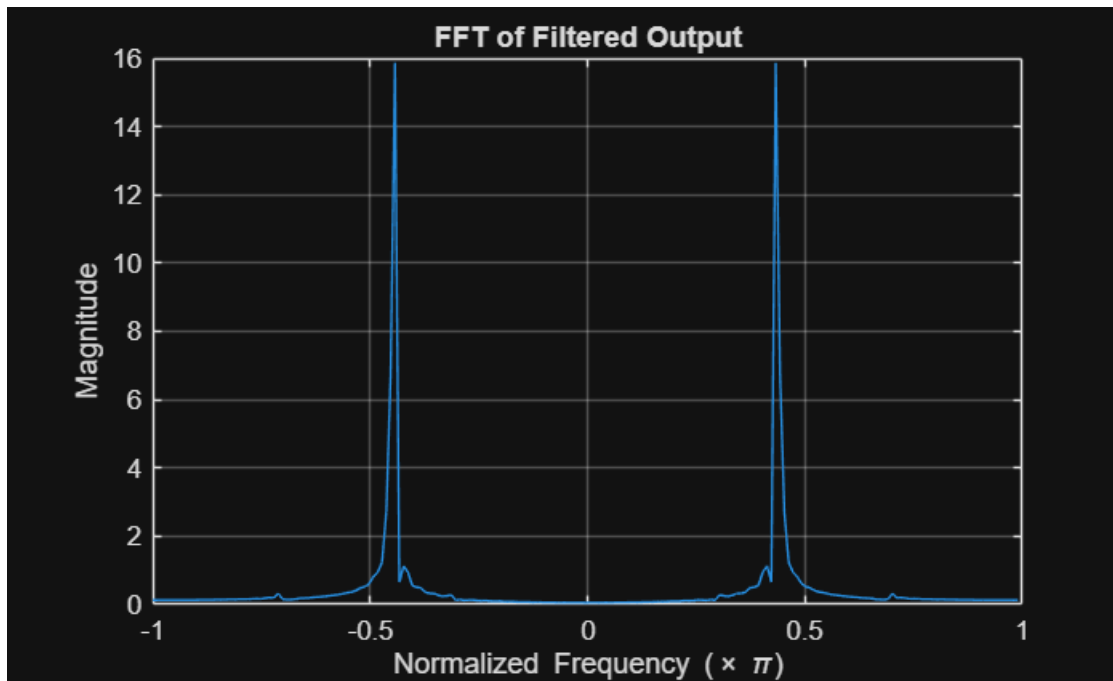
```
plot(f_axis, abs(Y_fft)/N_fft);
```

```
title('FFT of Filtered Output');
```

```
xlabel('Normalized Frequency (\times \pi)');
```

```
ylabel('Magnitude');
```

```
grid on;
```



%The FFT of the filtered output shows that the frequency component at 0.44π is preserved with high magnitude, while the components at 0.3π and 0.7π are significantly attenuated. This confirms that the designed filter effectively behaves as a band-pass filter centered at 0.44π .

%The attenuation of the unwanted frequency components satisfies the required condition of reducing them by at least a factor of 10, indicating that the chosen filter length is appropriate.

% Overall, the filter successfully isolates the desired frequency while suppressing noise and other undesired components, demonstrating proper band-pass filtering behavior.

Assessment Rubric

Lab 9: Frequency Response

Name:	Student ID:
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Marks distribution:

	Tasks	LR2 Code	LR5 Results	LR9 Report	AR4 Report Submission	AR7 Comments
In-Lab	Task 1	/8	/12	/8	/12	
	Task 2	/16	/8	/8		/8
Total =100		/24	/20	/16	/12	/8
CLO Mapped		CLO2	CLO2	CLO2	CLO3	CLO3

CLO – (LDL)	Total Points	Points Obtained
CLO2 – (Psy-4)	80	
CLO3 – (Aff-3)	20	
Total	100	